

# Chemistry

## Unit 3

### Practice Exam Answers

#### Section 1: Multiple choice

(32 marks)

##### Question 1

- C** There is no proton transfer so the reaction is not an acid–base one. There is a transfer of electrons from Zn to  $\text{H}^+$  to form  $\text{Zn}^{2+}$  and  $\text{H}_2$ . The Zn loses electrons so is oxidised by the  $\text{H}^+$ , which is therefore the oxidising agent.

##### Question 2

- B** Two species that differ only by one  $\text{H}^+$  form a conjugate acid–base pair.

##### Question 3

- C** The higher the  $K_a$ , the stronger the acid. So chlorous acid is stronger than nitrous acid. A strong acid releases a greater proportion of  $\text{H}^+$  ions than does a weaker acid. So, when concentrations are equal, the pH of the stronger acid will be lower.

##### Question 4

- D** Salts containing highly charged metal ions are hydrated and these hydrated ions can donate a proton to water to form an acidic solution. The  $\text{Cl}^-$  ions are such weak bases that they do not undergo hydrolysis, so aluminium chloride ( $\text{AlCl}_3$ ) forms an acidic solution. Potassium chloride forms a neutral solution because it is a salt of a strong acid and a strong base. The  $\text{Cl}^-$  ions are such weak bases that they do not undergo hydrolysis, nor do the  $\text{K}^+$  ions. Potassium carbonate forms a basic solution.  $\text{K}^+$  ions do not undergo hydrolysis but the  $\text{CO}_3^{2-}$  ions act as bases with water  $\text{CO}_3^{2-}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HCO}_3^-(\text{aq}) + \text{OH}^-(\text{aq})$ .

##### Question 5

- D** For a change of 1 unit in pH, the concentration of  $\text{H}^+(\text{aq})$  changes by a factor of 10. So a change of pH of 2 units means a change of  $\text{H}^+(\text{aq})$  concentration of 100. Because the pH of beaker A is higher, then the  $\text{H}^+(\text{aq})$  concentration will be 100 times lower.

##### Question 6

- A** To act as either a Brønsted–Lowry acid or base, a species must be able to either accept or donate a  $\text{H}^+(\text{aq})$ .  $\text{NH}_4^+$ , having a positive charge is unlikely to attract a  $\text{H}^+(\text{aq})$ .  $\text{CO}_3^{2-}(\text{aq})$  has no  $\text{H}^+(\text{aq})$  to donate.  $\text{CH}_4$  has no acidic hydrogens.

##### Question 7

- D** The reaction is exothermic ( $\Delta H < 0$ ) so an increase in temperature will favour the reverse reaction. So the value of the equilibrium constant will decrease and so will the equilibrium yield of  $\text{NO}(\text{g})$ .

### Question 8

- D** The number of mole of reactants in the equation is the same as the number of mole of product, so changing the volume of the mixture has no effect on the position of equilibrium or the amounts of reactants and products present.

However, as the volume of the mixture has been decreased, all concentrations will be increased. Because of the increase in concentrations, the rate of both forward and reverse reactions will increase by the same extent, so equilibrium is maintained.

### Question 9

- B** The direction of electron flow indicates that Mn is oxidised. The overall equation is  $\text{Mn(s)} + \text{Sn}^{4+}(\text{aq}) \rightarrow \text{Mn}^{2+}(\text{aq}) + \text{Sn}^{2+}(\text{aq})$ . So  $\text{Sn}^{4+}(\text{aq})$  is the oxidising agent.

### Question 10

- B** The half equation at the ring is  $\text{Au}^{3+}(\text{aq}) + 3\text{e}^- \rightarrow \text{Au(s)}$ . Because this is a reduction, it will take place at the cathode. In order to attract the positively charged  $\text{Au}^{3+}(\text{aq})$  ions, the ring needs to be attached to the negative terminal of the power supply.

### Question 11

- A** Let  $x$  be the oxidation number of P.

$$\text{In } \text{P}_4\text{O}_6, 4x - 12 = 0 \quad \text{so } x = +3$$

$$\text{In } \text{POCl}_3, x - 2 - 3 = 0 \quad \text{so } x = +5$$

$$\text{In } \text{H}_3\text{PO}_4, 3 + x - 8 = 0 \quad \text{so } x = +5$$

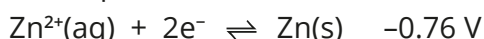
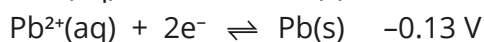
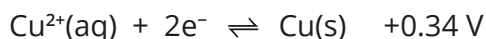
$$\text{Na}_2\text{HPO}_4, 2 + 1 + x - 8 = 0 \quad \text{so } x = +5$$

### Question 12

- B** The oxidation number of Mn in option B is the highest at +7.

### Question 13

- B** Consider the following standard reduction potentials.



If metal M reacts with 2 of the solutions but not 3, the standard reduction potential of  $\text{M}^{2+}(\text{aq}) + 2\text{e}^- \rightleftharpoons \text{M(s)}$  must be between  $-0.13 \text{ V}$  and  $+0.34 \text{ V}$ .

### Question 14

- C** From the reduction potential above, Zn is the strongest reducing agent.

### Question 15

- A**  $\text{Br}_2$  is a stronger oxidising agent or more reactive than  $\text{I}_2$  but a weaker oxidising agent or less reactive than  $\text{Cl}_2$ . So bromine will displace iodine from solution but will not displace chlorine.

### Question 16

- A** Because the temperature is constant, the value of the equilibrium constant will remain the same.

The addition of  $\text{OH}^-$  ions will cause the equilibrium to shift to the left of the equation to partly compensate for the added  $\text{OH}^-$  ions. So, at the new equilibrium, the concentration of  $\text{NH}_4^+$  ions will be lower and that of  $\text{NH}_3$  will be higher. However, only some of the added  $\text{OH}^-$  ions have been consumed by the equilibrium shift, so at the new equilibrium there will be an overall higher concentration of  $\text{OH}^-$  ions and the pH will thus be higher.

### End of section 1

## Section 2: Short answer

(48 marks)

(Note: In multi-mark questions each allocated mark is indicated as \* and total marks are indicated in parenthesis.)

### Question 17

a

| Equation (2 marks)                                                                                                | Explanation (2 marks)                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{HCOOH(aq)} + \text{H}_2\text{O(l)} \rightarrow \text{HCOO}^-\text{(aq)} + \text{H}_3\text{O}^+\text{(aq)}$ | HCOOH has donated a proton ( $\text{H}^+\text{(aq)}$ ) to water so HCOOH is acting as an acid.<br>$\text{H}_2\text{O(l)}$ has accepted a proton ( $\text{H}^+\text{(aq)}$ ) from HCOOH so water is acting as a base. |
| $\text{NH}_3\text{(g)} + \text{HCl(g)} \rightarrow \text{NH}_4\text{Cl(s)}$                                       | HCl has donated a proton ( $\text{H}^+$ ) to $\text{NH}_3$ so HCOOH is acting as an acid.<br>$\text{NH}_3$ has accepted a proton ( $\text{H}^+$ ) from HCl so $\text{NH}_3$ is acting as a base.                     |

b

| Equation (2 marks)                                                                                         | Explanation (2 marks)                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{Mg(s)} + 2\text{H}^+\text{(aq)} \rightarrow \text{Mg}^{2+}\text{(aq)} + \text{H}_2\text{(g)}$       | Zn has lost $2\text{e}^-$ so it has been oxidised to $\text{Zn}^{2+}$ .<br>Two $\text{H}^+\text{(aq)}$ have gained $\text{e}^-$ so they have been reduced to $\text{H}_2$ .                                                                                                                                                                    |
| $\text{CH}_4\text{(g)} + 2\text{O}_2\text{(g)} \rightarrow \text{CO}_2\text{(g)} + 2\text{H}_2\text{O(g)}$ | The oxidation number of C in $\text{CH}_4\text{(g)}$ is $-4$ and in $\text{CO}_2\text{(g)}$ it is $+4$ . As the oxidation number has increased, the carbon has been oxidised.<br>The oxidation number of O in $\text{O}_2$ is $0$ and in $\text{H}_2\text{O}$ it is $-2$ . As the oxidation number has decreased, the oxygen has been reduced. |

### Question 18

- a  $\text{CH}_3\text{COOH(aq)} + \text{H}_2\text{O(l)} \rightleftharpoons \text{CH}_3\text{COO}^-\text{(aq)} + \text{H}_3\text{O}^+\text{(aq)}$  (1 mark)
- b i If a small amount of HCl is added, this increases the concentration of  $\text{H}_3\text{O}^+\text{(aq)}$  in solution. The equilibrium shifts to the left, using up some of the added  $\text{H}_3\text{O}^+\text{(aq)}$  and so there is little change in the pH of the solution. (1 mark)
- ii If a small amount of NaOH is added, this releases  $\text{OH}^-$  ions which react with  $\text{H}_3\text{O}^+\text{(aq)}$  in solution\*. So the concentration of  $\text{H}_3\text{O}^+\text{(aq)}$  decreases. The equilibrium shifts to the right, replacing some of the reacted  $\text{H}_3\text{O}^+\text{(aq)}$  and so there is little change in the pH of the solution. (2 marks)
- c a buffer solution (1 mark)

### Question 19

- a  $\text{HCO}_3^-\text{(aq)} + \text{H}_2\text{O(l)} \rightarrow \text{H}_2\text{CO}_3\text{(aq)} + \text{OH}^-\text{(aq)}$  (1 mark)
- $\text{HCO}_3^-\text{(aq)} + \text{H}_2\text{O(l)} \rightarrow \text{CO}_3^{2-}\text{(aq)} + \text{H}_3\text{O}^+\text{(aq)}$  (1 mark)
- b If the pH is 8, it means that  $[\text{H}_3\text{O}^+\text{(aq)}] < [\text{OH}^-]^*$   
So the first equation must occur to the greater extent\*. (2 marks)

### Question 20

(5 marks)

|                                 | Higher for nitric acid | Higher for nitrous acid | Same for both |
|---------------------------------|------------------------|-------------------------|---------------|
| pH                              |                        | ✓                       |               |
| concentration of hydronium ions | ✓                      |                         |               |
| amount of hydroxide ions        |                        | ✓                       |               |
| $K_a$ value                     | ✓                      |                         |               |
| $K_w$ value                     |                        |                         | ✓             |

### Question 21

a

(3 marks)

|               | Should be rinsed with ...      |
|---------------|--------------------------------|
| conical flask | distilled water                |
| pipette       | the sodium carbonate solution  |
| burette       | the hydrochloric acid solution |

b An acid–base indicator is a weak acid that has a distinctly different colour from its conjugate base.

(1 mark)

c

(2 marks)

|                  |                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------|
| systematic error | e.g. incorrectly recording the concentration of sodium carbonate; rinsing pipette with water                     |
| random error     | e.g. estimating level of acid between gradations of burette; gauging colour change of indicator at the end point |

### Question 22

(3 marks)

| Equation                                                                                                                | Equilibrium law expression                                                              |
|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + 3\text{H}_2(\text{g})$   | $\frac{[\text{CO}] \times [\text{H}_2]^3}{[\text{CH}_4] \times [\text{H}_2\text{O}]}$ * |
| $\text{Ag}^+(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \rightleftharpoons \text{Ag}(\text{s}) + \text{Fe}^{3+}(\text{aq})$ | $\frac{[\text{Fe}^{3+}]}{[\text{Ag}^+] \times [\text{Fe}^{2+}]}$ *                      |
| $2\text{NO}(\text{g}) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{NOCl}(\text{g})$ *                              | $K = \frac{[\text{NOCl}]^2}{[\text{NO}]^2[\text{Cl}_2]}$                                |

### Question 23

a bromine in aqueous solution

(1 mark)

b (2 marks for all three correct colour changes in column 1; 1 mark for two correct colour changes in column 1; 2 marks for each correct explanation)

(8 marks)

| Change                                                | Intensity of orange colour of mixture increased/ decreased/the same | Explanation                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The mixture is heated.                                | becomes lighter orange*                                             | The forward reaction is exothermic. Increasing the temperature increases the rate of both forward and reverse reactions but the rate of the endothermic reaction (reverse reaction here) is increased more*. So the equilibrium shifts to the left and less orange bromine is present*.                                                                                             |
| A small amount of concentrated KBr solution is added. | becomes darker orange*                                              | At the instant the concentration of one of the reactants increases, the rate of the forward reaction increases and is therefore greater than the rate of the reverse reaction*. So at the new equilibrium position, more products have formed. More bromine causes the darker colour*.                                                                                              |
| A small amount of concentrated KOH is added.          | becomes lighter orange*                                             | The $\text{OH}^-$ ions added react with the $\text{H}^+(\text{aq})$ present. The concentration of one of the reactants decreases, so the rate of the forward reaction decreases and until equilibrium is re-established, it is less than the rate of the reverse reaction*. So at the new equilibrium position, less products have formed. Less bromine causes the lighter colour*. |

### Question 24

- a**  $\text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow \text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq}) + 2\text{e}^-$  (1 mark)
- b**  $\text{MnO}_4^-(\text{aq}) + 8\text{H}^+(\text{aq}) + 5\text{e}^- \rightarrow \text{Mn}^{2+}(\text{aq}) + 4\text{H}_2\text{O}(\text{l})$  (1 mark)
- c**  $2\text{MnO}_4^-(\text{aq}) + 5\text{SO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l}) \rightarrow 2\text{Mn}^{2+}(\text{aq}) + 5\text{SO}_4^{2-}(\text{aq}) + 4\text{H}^+(\text{aq})$  (1 mark)
- d** Let  $x$  be the oxidation number of S. (1 mark)
- $$x - 8 = -2 \quad \text{so } x = +6$$

### Question 25

- a** The blue becoming lighter indicates that the  $\text{Cu}^{2+}(\text{aq})$  ions are being reduced. (2 marks)
- So the cathode reaction is  $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})^*$
- At the anode, the reaction must be  $\text{M}(\text{s}) \rightarrow \text{M}^{2+}(\text{aq}) + 2\text{e}^-$
- So metal M causes the reduction of the  $\text{Cu}^{2+}(\text{aq})$  and therefore M is the stronger reducing agent\*.
- b** M is the reducing agent and  $\text{Cu}^{2+}(\text{aq})$  must be the oxidising agent. (2 marks)

$$E^\circ_{\text{cell}} = E^\circ_{\text{oxidant}} - E^\circ_{\text{reductant}}$$

$$1.52 = 0.34 \text{ (from data book)} - E^\circ_{\text{reductant}}^*$$

$$E^\circ_{\text{reductant}} = 0.34 - 1.52 = -1.18 \text{ V} = \text{standard reduction potential for the } \text{M}^{2+}/\text{M} \text{ half-cell}^*$$

End of section 2

### Section 3: Extended answer

(43 marks)

(Note: In multi-mark questions each allocated mark is indicated as \* and total marks are indicated in parentheses.)

#### Question 26

- a i  $[H^+] = 0.010 \text{ mol L}^{-1}$  so  $[OH^-] = \frac{10^{-14}}{0.010} = 10^{-12} \text{ mol L}^{-1}$  (1 mark)
- ii  $pH = -\log 0.010 = 2$  (1 mark)
- iii  $n(H^+) = 0.010 \times 0.020 = 0.00020$  (1 mark)
- b (6 marks)

| In comparison to 20.00 mL 0.010 mol L <sup>-1</sup> HCl,                |                                                                                                                                                                                                            |                                                                                                                                                                   |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                         | the pH of the solution is ...                                                                                                                                                                              | the amount of 0.010 mol L <sup>-1</sup> NaOH the solution would need for neutralisation is ...                                                                    |
| <b>40.00 mL 0.010 mol L<sup>-1</sup> HCl</b>                            | higher/the same/lower<br>Justification:<br>The pH depends on the concentration of H <sup>+</sup> (aq) in solution not its volume.                                                                          | higher/the same/lower<br>Justification:<br>The amount in mole of H <sup>+</sup> (aq) present is higher due to the larger volume of solution.                      |
| <b>20.00 mL 0.010 mol L<sup>-1</sup> of the weak acid HOCl</b>          | higher/the same/lower<br>Justification:<br>HOCl is hydrolysed to a lesser extent and so will release fewer H <sup>+</sup> (aq). Lower concentration of H <sup>+</sup> (aq) results in a higher pH.         | higher/the same/lower<br>Justification:<br>NaOH reacts with all the acid molecules. The amount in mol of each acid is the same.                                   |
| <b>20.00 mL of 0.010 mol L<sup>-1</sup> H<sub>2</sub>SO<sub>4</sub></b> | higher/the same/lower<br>Justification:<br>H <sub>2</sub> SO <sub>4</sub> is a diprotic acid and so will release more than one H <sup>+</sup> (aq) in solution. More H <sup>+</sup> (aq) means a lower pH. | higher/the same/lower<br>Justification:<br>Twice as many OH <sup>-</sup> ions are required to neutralise the 2 acidic protons in H <sub>2</sub> SO <sub>4</sub> . |

- c i  $H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$  (1 mark)
- ii In order to obtain sufficient concordant results. The first titration is usually done quickly and so is prone to error. (1 mark)
- iii average volume of HCl used =  $\frac{17.30 + 17.35 + 17.25}{3} = 17.30 \text{ mL}$  (result of first titration discarded)
- $n(HCl) = 0.100 \times 0.01730 = 0.001730 = n(NaOH)^*$
- $c(KOH) = \frac{0.001730}{0.02000} = 0.0865 \text{ mol L}^{-1}$  (3 sig. figs)\* (2 marks)
- iv Student B is correct. Any of the listed indicators would be suitable\* as in this titration between a strong acid and a strong base, the pH changes dramatically with the addition of a drop of acid near equivalence point. All of these indicators would change colour with the addition of that single drop\*. (2 marks)

#### Question 27

- a i (4 marks)

at cathode  
 $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)^*$

at anode  
 $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^- *$   
 $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^- *$   
 $Pb(s) \rightarrow Pb^{2+}(aq) + 2e^- *$

- ii As most of the metal in the anode oxidises, the impurities remain unreacted because they are weak reducing agents\* and drop to the bottom of the tank\*. (2 marks)
  - iii  $\text{Zn}^{2+}(\text{aq})$  and  $\text{Pb}^{2+}(\text{aq})$ , which are formed at the anode, are weaker oxidising agents than  $\text{Cu}^{2+}(\text{aq})$ \* and so will not be reduced at the cathode but remain in solution\*. (2 marks)
  - iv The colour is due to  $\text{Cu}^{2+}(\text{aq})$ . The number of  $\text{Cu}^{2+}(\text{aq})$  produced at the anode is equal to the number of  $\text{Cu}^{2+}(\text{aq})$  reacting at the cathode. So the concentration of  $\text{Cu}^{2+}(\text{aq})$  in solution remains constant and so does the colour of the solution. (1 mark)
  - v According to the equation  $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \rightarrow \text{Cu}(\text{s})$ , 1 mole of electrons will result in 0.50 mol of  $\text{Cu}^*$ . So  $m(\text{Cu})$  deposited =  $0.50 \times 63.55 = 31.8 \text{ g}^*$  (2 marks)
- b  $\text{Cu}^{2+}(\text{aq})$  ions are stronger oxidising agents than water. (3 marks)

$\text{Al}^{3+}(\text{aq})$  ions are weaker oxidising agents than water\*.

So in aqueous solution,  $\text{Cu}^{2+}(\text{aq})$  will be reduced in preference to water but water will be reduced in preference to  $\text{Al}^{3+}(\text{aq})$  ions\*.

In a molten aluminium compound, the only oxidising agent present is the  $\text{Al}^{3+}$  which is reduced electrolytically\*.

### Question 28

- a  $\frac{[\text{PCl}_5]}{[\text{PCl}_3][\text{Cl}_2]}$  or  $\frac{pp\text{PCl}_5}{pp\text{PCl}_3 \cdot pp\text{Cl}_2}$  (1 mark)
- b The concentration of  $\text{PCl}_3$  dropped from 1 to almost 0.1 indicating that the reaction proceeded to a large extent\*. So the equilibrium constant, at that temperature, is  $> 1^*$ . (2 marks)
- c i As the temperature increased, the amount of product decreased and the amount of leftover reactant increased. (That is, the rate of the forward reaction increased by a lesser amount than the rate of the reverse reaction.)\* So the forward reaction must be exothermic\*. (2 marks)
- ii At  $325^\circ\text{C}$ , the amount of product has decreased and the leftover reactant increased\* so the value of the equilibrium constant must be smaller\* at that temperature than at  $225^\circ\text{C}$ . (2 marks)
- d i The pressure on the equilibrium mixture was increased by reducing the volume of the equilibrium mixture. (1 mark)
- ii As the pressure was increased, the system readjusted itself in order to reduce the pressure. So the equilibrium readjusted to favour the side of the equation with fewer moles—in this case  $\text{PCl}_5$ . (1 mark)
- iii The shift in equilibrium only partially compensated for the decrease in volume/increase in pressure so the net effect is an increase in the concentrations of all the components of the equilibrium system. (1 mark)
- e A curve starting at 1.0 and coming to equilibrium at 0.1 at a point before the 20-second mark. (1 mark)
- f As the temperature is increased, the average kinetic energy of molecules increases and so a greater proportion of molecules have energy greater than or equal to the activation energy for the reaction\*. Therefore, a much higher proportion of molecular collisions will successfully result in a reaction\*.
- A smaller contributor to the increased rate is that molecules will collide more frequently at the higher temperature\*. (3 marks)

### End of answers